Ideal formulations

Outline

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Learning objectives

At the end of this chapter you will be able to:

- compute the convex hull of the feasible solutions of a small ILP problem, and classify its points as vertices, boundary points or interior points;
- define a formulation of an ILP problem, and decide whether it is ideal;
- explain why an LP problem over an ideal formulation solves the ILP problem;
- compare two formulations of the same problem through their LP relaxations;
- explain why ideal formulations exist but are hard to find.

Prerequisites

- From this module, the chapters «Integer linear programming» and «Primal and dual bounds»: the feasible region of an ILP problem, the LP relaxation and its bound, and the car example
- From the module *Linear Programming*, Chapter 15 «The geometry of Linear Programming»: vertices of a polyhedron, and the fact that an LP problem with an optimal solution has an optimal vertex
- From the module *Convex optimization*, Chapter 1 «Convex sets»: convex combinations and convex hulls (a short recap is given in this chapter)

Where we were

- In the car example we have the lower bound 14 and the upper bound $\lfloor z(\mathrm{LP}) \rfloor = 15$
- The gap of $\frac{1}{14}$ is due to the polyhedron of the constraints, which contains fractional points outside the convex hull of the integer points
- This chapter defines the polyhedron that gives **no gap**

Why ideal formulations?

Introduction: Why ideal formulations?

- The upper bound of the LP relaxation depends on the polyhedron that we write
- We ask which polyhedron gives the best bound
- The answer is the guiding idea of the whole chapter

Many systems, many bounds

- The same set of integer points can be described by **many different systems** of linear inequalities
- Each of them gives a different LP relaxation, with a different bound

The best polyhedron

The best polyhedron is the **smallest one**
that contains all the integer points.

If we could use it, the LP relaxation
would return an integer solution,
and the ILP problem would be as easy as an LP problem.

Convex hull and ideal formulation

- This polyhedron is the **convex hull** of the feasible solutions
- A system of inequalities that describes it is an **ideal formulation**

Summary

- Different systems of inequalities for the same integer points give different bounds
- The smallest polyhedron containing the integer points, the **convex hull**, makes the LP relaxation return an integer solution
- A system that describes it is an **ideal formulation**

The convex hull

Introduction: The convex hull

- We recall convex combinations and the convex hull of a set of points
- We see that the convex hull of a finite set is a polytope whose vertices are points of the set
- We compute the convex hull of the car example, and classify its 16 integer points

Definition 1: convex hull

Given a set $S \subseteq \mathbb{R}^n$, *a* ***convex combination*** *of points* $\boldsymbol{s}^1, \boldsymbol{s}^2, \ldots, \boldsymbol{s}^k \in S$ *is a point*

$$\sum_{h=1}^{k} \lambda_h \boldsymbol{s}^h \qquad \text{with} \qquad \lambda_h \geq 0 \quad \text{and} \quad \sum_{h=1}^{k} \lambda_h = 1.$$

The ***convex hull*** *of* S, *denoted by* $\mathrm{conv}(S)$, *is the set of all the convex combinations of points of* S.

Properties of the convex hull

- The convex hull is the **smallest convex set** containing S
- If S is a finite set of points, the convex hull is a bounded polyhedron (a **polytope**), and its **vertices** are points of S
- The other points of S lie either on the **boundary** of the polytope, or in its **interior**: they are not needed to describe it

The convex hull of the feasible solutions

- For an ILP problem we take $S = F$, the set of the feasible solutions
- The vertices of $\mathrm{conv}(F)$ are **integer feasible solutions**

We now compute $\mathrm{conv}(F)$ for the car example, and look at where its 16 integer points lie.

Example (1/3)

- The convex hull of the 16 integer points of the car example has 6 **vertices**:

$$(0,0), \quad (4,0), \quad (4,1), \quad (3,2), \quad (1,3), \quad (0,3).$$

Example (2/3)

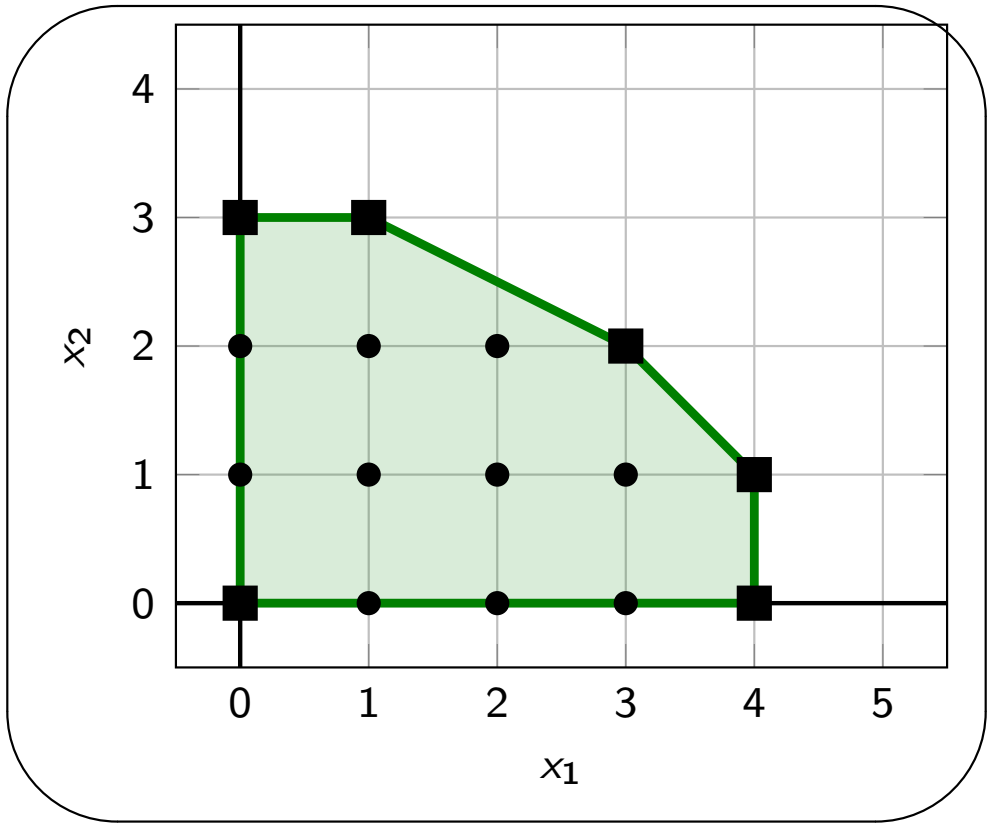
- Of the other 10 points, five lie on its **boundary**, namely

$$(0,1), \quad (0,2), \quad (1,0), \quad (2,0), \quad (3,0)$$

- and five in its **interior**, namely

$$(1,1), \quad (2,1), \quad (3,1), \quad (1,2), \quad (2,2)$$

Example (3/3)



The convex hull of the 16 integer points of the car example (green). Its 6 vertices (black squares) are integer points; five other points lie on its boundary and five in its interior.

Summary

- $\mathrm{conv}(S)$ is the set of the convex combinations of points of S: the smallest convex set containing S
- For a finite set S it is a **polytope**, whose vertices are points of S
- In the car example, $\mathrm{conv}(F)$ has 6 vertices; of the other 10 points, 5 lie on the boundary and 5 in the interior

Formulations and ideal formulations

Introduction: Formulations and ideal formulations

- We define a formulation: a polyhedron with exactly the integer points of the ILP problem
- We define the ideal formulation, the one equal to the convex hull
- We prove that, with an ideal formulation, the LP relaxation solves the ILP problem
- On the car example: the original formulation is not ideal, and we write an ideal one

Definition 2: formulation

*A **formulation** of an ILP problem with feasible region F is a polyhedron*

$$P = \left\{ \boldsymbol{x} \in \mathbb{R}^n : \sum_j a_{ij}\, x_j \leq b_i,\ \forall i, \quad x_j \geq 0,\ \forall j \right\}$$

such that

$$P \cap \mathbb{Z}^n = F.$$

*The **LP relaxation of the formulation** is the problem of maximizing $\sum_j c_j\, x_j$ over P.*

Every formulation contains the convex hull

- An ILP problem has **many formulations**: they share the integer points and differ in the fractional points
- Since P is convex and contains F, every formulation contains the convex hull:

$$\text{conv}(F) \subseteq P.$$

Among all the formulations,
the smallest possible one is $\mathrm{conv}(F)$ itself:
we give it a name.

Definition 3: ideal formulation

A formulation P of an ILP problem is ***ideal*** *if*

$$P = \text{conv}(F),$$

that is, if the feasible region of its LP relaxation is exactly the convex hull of the integer feasible solutions.

What do we gain with an ideal formulation?
The next theorem says that the LP relaxation
is enough to solve the ILP problem.

Theorem 1: LP over an ideal formulation

Let P be an ideal formulation of an ILP problem with $F \neq \emptyset$ and P bounded. Then

- *the LP relaxation has an optimal vertex,*
- *every optimal vertex is an optimal solution of the ILP problem, and*
- $z(\mathrm{LP}) = z(\mathrm{ILP})$.

- Idea of the proof: an optimal vertex of $\mathrm{conv}(F)$ is a point of F, and no ILP solution can beat the LP bound

Proof

- The polyhedron $P = \mathrm{conv}(F)$ is non-empty and bounded, so the LP problem has an optimal vertex of P
- The vertices of $\mathrm{conv}(F)$ belong to F, so this vertex is a feasible ILP solution, whose value is $z(\mathrm{LP})$
- Since
$$z(\mathrm{ILP}) \leq z(\mathrm{LP}),$$
the vertex is optimal for the ILP problem

□

Why an optimal vertex

- The simplex algorithm returns an optimal vertex, so with an ideal formulation it returns an **integer solution**
- If the LP problem has several optimal solutions, a method that does not return vertices can give a fractional point on the optimal face
- This is why the theorem speaks of an **optimal vertex**

Solving an ILP problem as an LP problem

With an ideal formulation,
an ILP problem is solved as an LP problem:
an **optimal vertex** of the LP relaxation
is an **optimal ILP solution**.

Is the formulation of the car example ideal?
We compare its polyhedron P with $\mathrm{conv}(F)$.

Example: a formulation that is not ideal (1/2)

- The point $(\frac{7}{2}, \frac{5}{2})$ belongs to the polyhedron P of the car example
- It does not belong to $\mathrm{conv}(F)$, because it violates

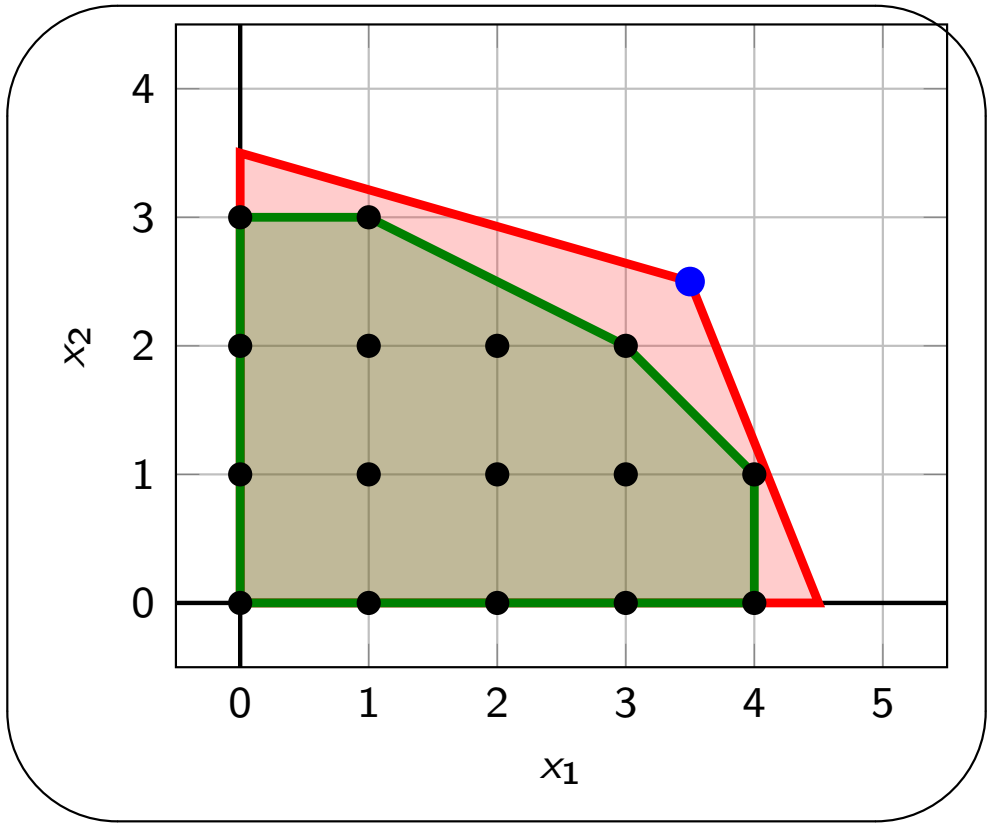
$$x_1 + x_2 \leq 5,$$

  which all the vertices of the hull of the car example satisfy

- Hence

$$\mathrm{conv}(F) \subsetneq P$$

Example: a formulation that is not ideal (2/2)



The polyhedron of the LP relaxation (red) contains the convex hull of the integer points (green), and the optimal solution of the LP relaxation (blue) is outside the hull.

If we describe $\mathrm{conv}(F)$ with linear inequalities,
we obtain an ideal formulation of the car example.

Example: an ideal formulation (1/3)

- The convex hull of the car example is described by the following four inequalities and the non-negativity constraints:

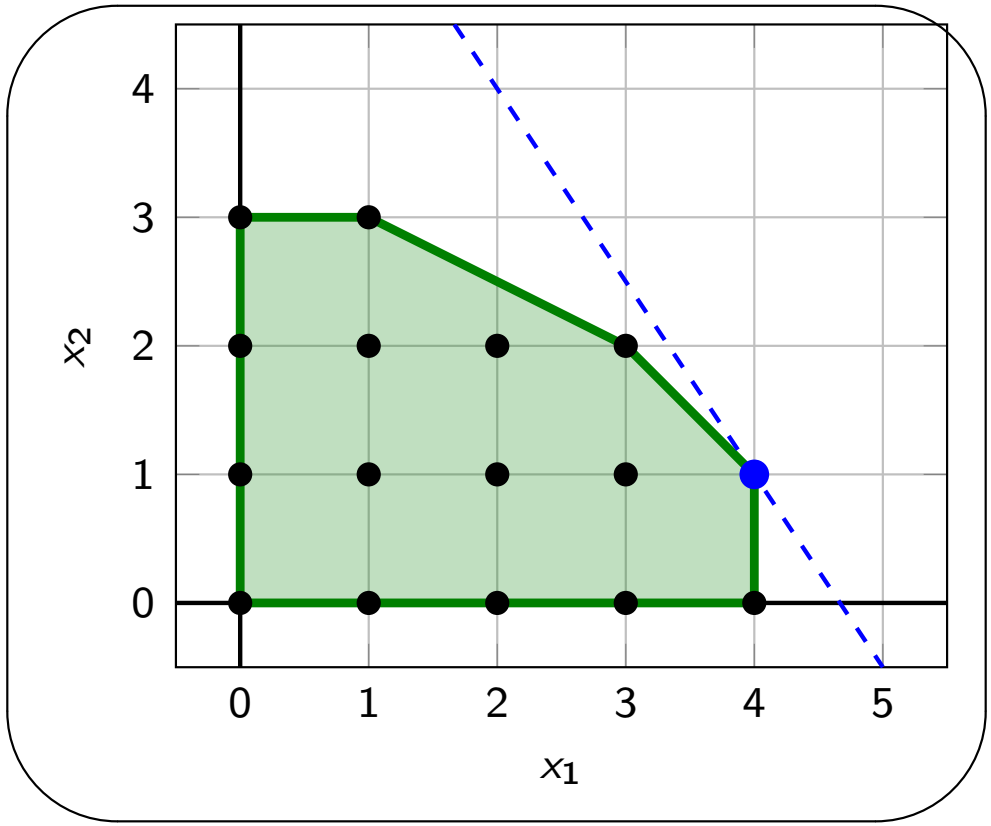
$$
\begin{array}{rrcrcl}
\max & 3\,x_1 & + & 2\,x_2 & & \\
\text{subject to} & x_1 & + & x_2 & \leq & 5 \\
 & x_1 & + & 2\,x_2 & \leq & 7 \\
 & x_1 & & & \leq & 4 \\
 & & & x_2 & \leq & 3 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$

Example: an ideal formulation (2/3)

- These inequalities define exactly the 6 vertices of the hull, and the same 16 integer points
- The LP relaxation of this **ideal formulation** has the optimal solution $(4, 1)$, which is integer, with

$$z(\mathrm{LP}) = 14 = z(\mathrm{ILP})$$

Example: an ideal formulation (3/3)



The ideal formulation of the car example: its feasible region (green) is the convex hull of the integer points. The optimal solution of the LP problem is the integer point $(4, 1)$ (blue).

Summary

Formulations:

- A polyhedron P with $P \cap \mathbb{Z}^n = F$; every formulation satisfies $\mathrm{conv}(F) \subseteq P$
- A formulation is **ideal** if $P = \mathrm{conv}(F)$

LP over an ideal formulation:

- An optimal vertex of the LP relaxation is an optimal ILP solution, and $z(\mathrm{LP}) = z(\mathrm{ILP})$
- Car example: the original P is not ideal, the ideal formulation has 4 inequalities and gives $(4, 1)$ with $z(\mathrm{LP}) = 14$

Comparing formulations

Introduction: Comparing formulations

- The same ILP problem has many formulations: which one should we use?
- We compare two formulations by inclusion of their polyhedra
- A smaller polyhedron gives a better upper bound: the ideal formulation is the best of all

Definition 4: stronger formulation

Given two formulations P_1 and P_2 of the same ILP problem, P_1 is ***stronger*** *than P_2 if*

$$P_1 \subseteq P_2.$$

A stronger formulation gives a better bound

- A stronger formulation never gives a worse upper bound
- The maximum over a smaller set is not larger, so

$$P_1 \subseteq P_2 \quad \Longrightarrow \quad z(\mathrm{ILP}) \;\leq\; z(\mathrm{LP}_1) \;\leq\; z(\mathrm{LP}_2).$$

The ideal formulation is the strongest

- The ideal formulation is the **strongest of all**, because it is contained in every formulation
- In the car example:
  - the original formulation gives the bound $\lfloor \frac{31}{2} \rfloor = 15$
  - the ideal formulation of the example above gives the bound 14

Which formulation to prefer

Among the formulations of the same ILP problem,
we prefer the ones with the **smallest polyhedron**:
their LP relaxations give the best upper bounds,
and the ideal formulation gives the optimal value.

Summary

- P_1 is **stronger** than P_2 if $P_1 \subseteq P_2$
- Then $z(\mathrm{ILP}) \leq z(\mathrm{LP}_1) \leq z(\mathrm{LP}_2)$: a stronger formulation never gives a worse bound
- The ideal formulation is the strongest: in the car example it improves the bound from 15 to 14

Existence of ideal formulations

Introduction: Existence of ideal formulations

- An ideal formulation would solve the ILP problem as an LP problem: does it always exist?
- We see when it exists, and why it is hard to find
- This leads to what we do in practice: valid inequalities

Ideal formulations exist

- An ideal formulation exists for every ILP problem whose feasible region is **bounded** or, more in general, whose data are **rational** (Meyer's theorem; in this module the data are integer)
- The convex hull of the integer points is a polyhedron, so it is described by a **finite number** of linear inequalities

Ideal formulations are hard to find

- Finding the ideal formulation is in general **very difficult**
- It can require an **exponential number** of inequalities
- Finding the necessary ones is often as hard as solving the ILP problem itself

What we do in practice

- In practice we keep the formulation we have and add, step by step, inequalities that
  - are satisfied by all the integer points
  - cut off fractional points

- They are the **valid inequalities** of the next chapter

Summary

- Ideal formulations exist for bounded feasible regions and, in general, for rational data
- They can need exponentially many inequalities, and finding them is often as hard as the ILP problem
- In practice we strengthen the formulation step by step with **valid inequalities**

Typical mistakes

Introduction: Typical mistakes

- We list the mistakes that are made most often with convex hulls and formulations
- They concern the points of the hull, how to check that a formulation is ideal, and how to strengthen a formulation
- Knowing them helps to read and solve the exercises correctly

Typical mistakes

Observation 1: Typical mistakes (1/2)

- ***Believing that the points that are not vertices lie strictly inside the hull.*** *They can lie on its boundary, as* $(0, 1)$ *and* $(3, 0)$ *in the car example.*
- ***Checking only the optimal solution.*** *A formulation can return an integer optimal solution without being ideal: the LP optimum can be integer for one objective function and fractional for another one.*

Typical mistakes

Observation: Typical mistakes (2/2)

- ***Believing that every formulation is ideal.*** *The formulation of the car example has the right integer points but contains fractional points outside the hull.*
- ***Adding inequalities that remove integer points.*** *A formulation must keep all the points of F: the polyhedron can shrink only down to the convex hull, never below it.*

Summary

Typical mistakes:

- Points that are not vertices can lie on the boundary of the hull
- An integer LP optimum for one objective function does not make a formulation ideal
- Not every formulation is ideal, and a formulation must never lose integer points

In summary

Introduction: In summary

- We collect the key ideas of the chapter
- We close with the dashboard of the car example: the bounds found so far

Summary (1/3)

- The convex hull of the feasible solutions is the smallest polyhedron containing them, and its vertices are integer feasible solutions
- A formulation is ideal if its polyhedron is the convex hull; then an optimal vertex of the LP relaxation is an optimal ILP solution
- A stronger formulation, with a smaller polyhedron, never gives a worse bound

Summary (2/3)

- The car example has an ideal formulation with 4 inequalities, which gives $z(\mathrm{LP}) = 14 = z(\mathrm{ILP})$
- Ideal formulations exist for rational data, but in general they need exponentially many inequalities and are hard to find

Summary (3/3)

Dashboard of the car example. The bounds on $z(\mathrm{ILP})$ found so far in this module:

Chapter	Lower bound	Upper bound	From
Primal and dual bounds	14	15	heuristic solution $(4,1)$, and $\lfloor z(\mathrm{LP}) \rfloor$ with $z(\mathrm{LP}) = \frac{31}{2}$
Ideal formulations	14	14	ideal formulation, optimal vertex $(4,1)$